

# More NP-complete Problems

**Theorem:** (proven in previous class)

**If:** Language  $A$  is NP-complete

Language  $B$  is in NP

$A$  is polynomial time reducible to  $B$

**Then:**  $B$  is NP-complete

Using the previous theorem,  
we will prove that 2 problems  
are NP-complete:

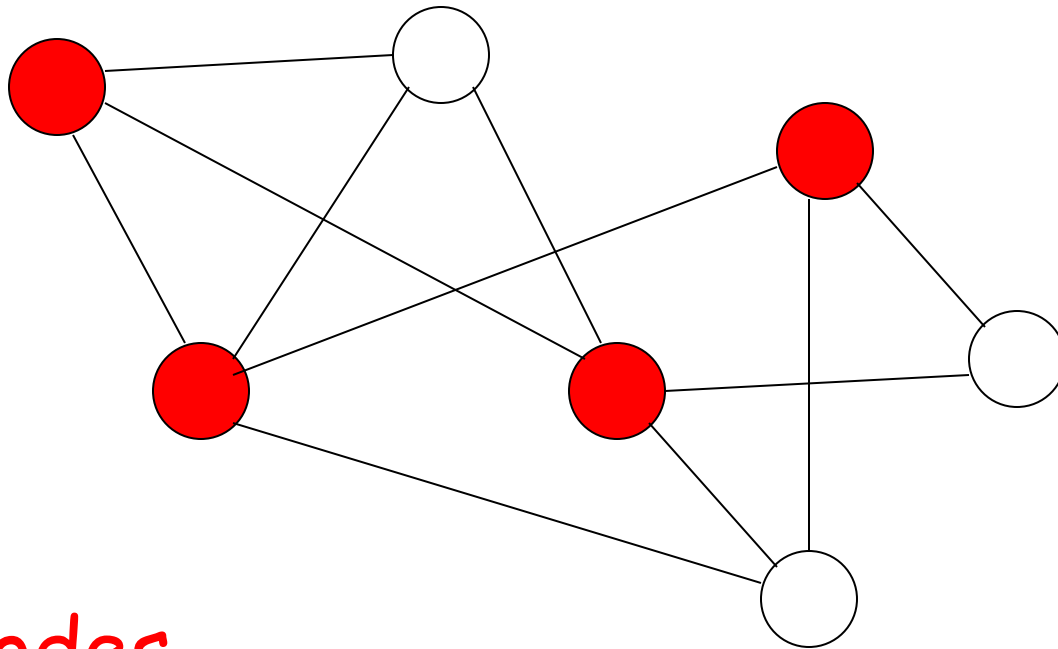
Vertex-Cover

Hamiltonian-Path

# Vertex Cover

Vertex cover of a graph  
is a subset of nodes  $S$  such that every edge  
in the graph touches one node in  $S$

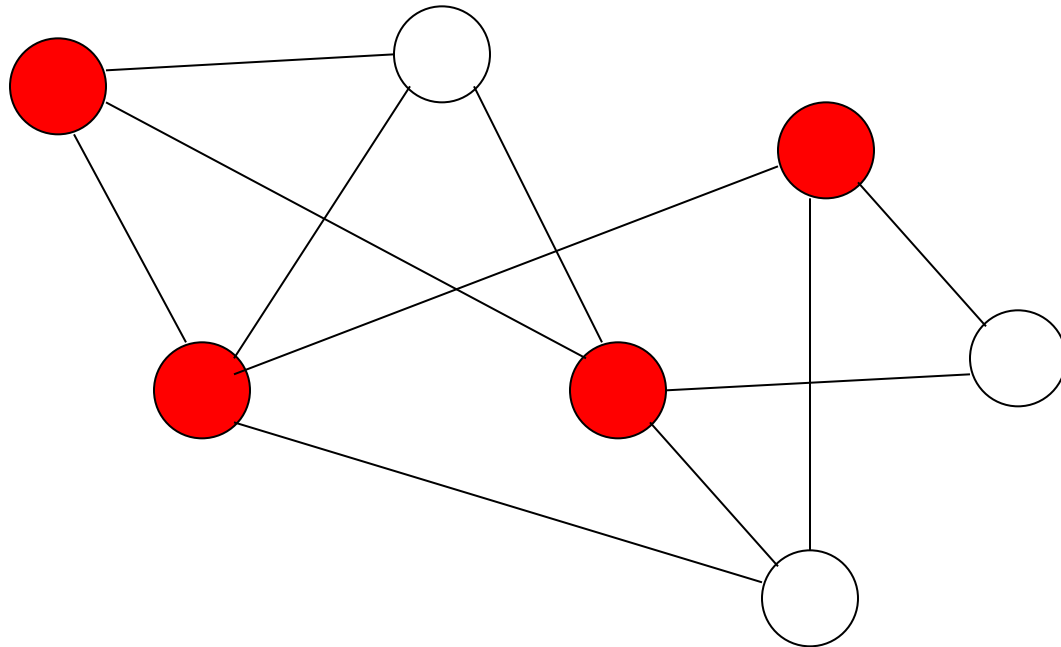
Example:



$S$  = red nodes

Size of vertex-cover  
is the number of nodes in the cover

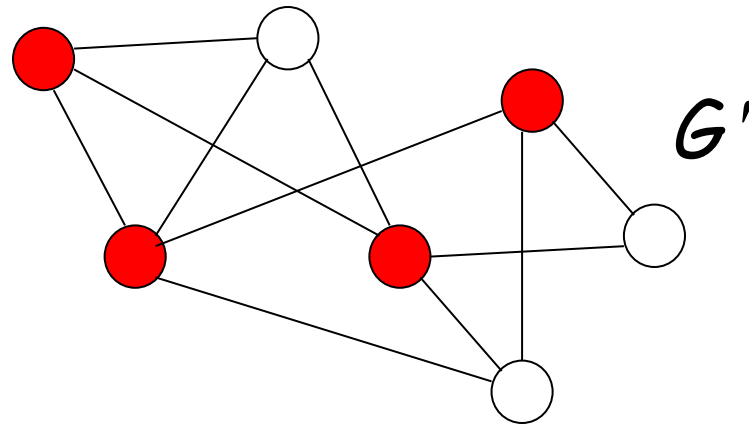
Example:  $|S|=4$



Corresponding language:

$\text{VERTEX-COVER} = \{ \langle G, k \rangle : \text{graph } G \text{ contains a vertex cover of size } k \}$

Example:



$\langle G', 4 \rangle \in \text{VERTEX-COVER}$

**Theorem:** VERTEX-COVER is NP-complete

**Proof:**

1. VERTEX-COVER is in NP

Can be easily proven

2. We will reduce in polynomial time

3CNF-SAT to VERTEX-COVER  
(NP-complete)

Let  $\varphi$  be a 3CNF formula  
with  $m$  variables  
and  $l$  clauses

Example:

$$\varphi = (\underbrace{x_1 \vee x_2 \vee x_3}_{\text{Clause 1}}) \wedge (\underbrace{\overline{x_1} \vee \overline{x_2} \vee \overline{x_4}}_{\text{Clause 2}}) \wedge (\underbrace{\overline{x_1} \vee x_3 \vee x_4}_{\text{Clause 3}})$$

$$m = 4$$

$$l = 3$$



Formula  $\varphi$  can be converted  
to a graph  $G$  such that:

$\varphi$  is satisfied

if and only if

$G$  Contains a vertex cover  
of size  $k = m + 2l$

$$\varphi = (x_1 \vee x_2 \vee x_3) \wedge (\overline{x_1} \vee \overline{x_2} \vee \overline{x_4}) \wedge (\overline{x_1} \vee \overline{x_3} \vee x_4)$$

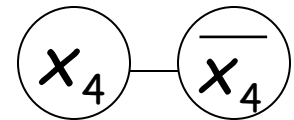
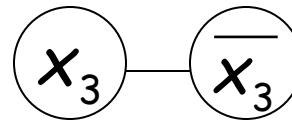
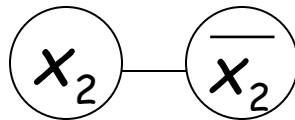
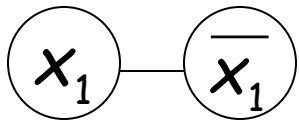
Clause 1

Clause 2

Clause 3

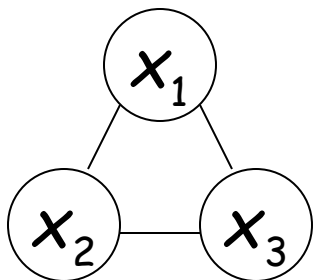
Variable Gadgets

$2m$  nodes

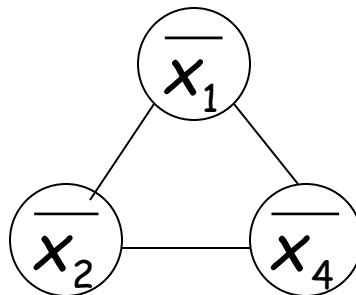


Clause Gadgets

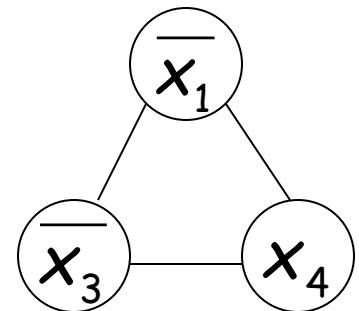
$3$  nodes



Clause 1

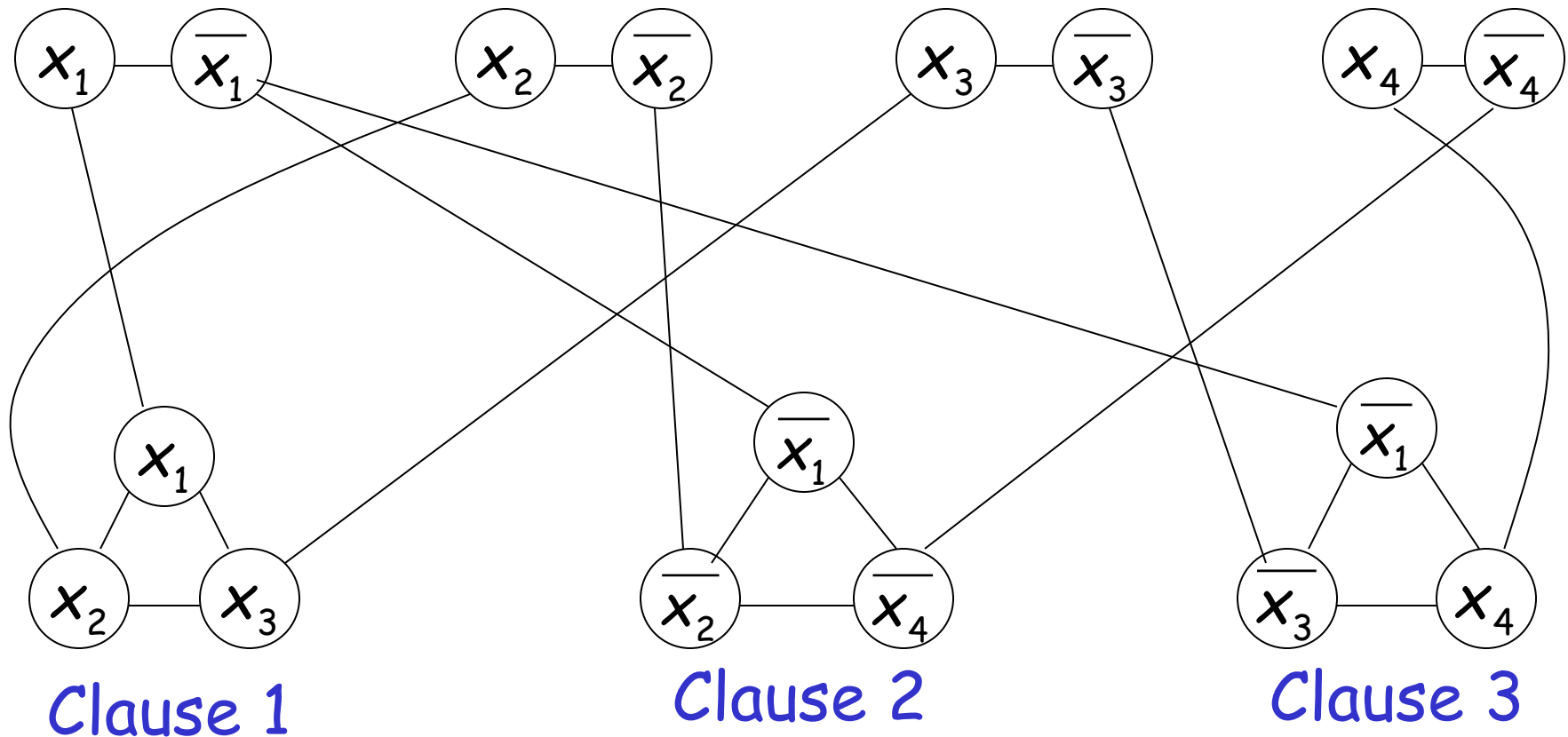


Clause 2



Clause 3

$$\varphi = (\underbrace{x_1 \vee x_2 \vee x_3}_{\text{Clause 1}}) \wedge (\underbrace{\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_4}_{\text{Clause 2}}) \wedge (\underbrace{\bar{x}_1 \vee \bar{x}_3 \vee x_4}_{\text{Clause 3}})$$



First direction in proof:

If  $\varphi$  is satisfied,

then  $G$  contains a vertex cover of size

$$k = m + 2l$$

Example:

$$\varphi = (x_1 \vee x_2 \vee x_3) \wedge (\overline{x_1} \vee \overline{x_2} \vee \overline{x_4}) \wedge (\overline{x_1} \vee \overline{x_3} \vee x_4)$$

Satisfying assignment

$$x_1 = 1 \quad x_2 = 0 \quad x_3 = 0 \quad x_4 = 1$$

We will show that  $G$  contains  
a vertex cover of size

$$k = m + 2l = 4 + 2 \cdot 3 = 10$$

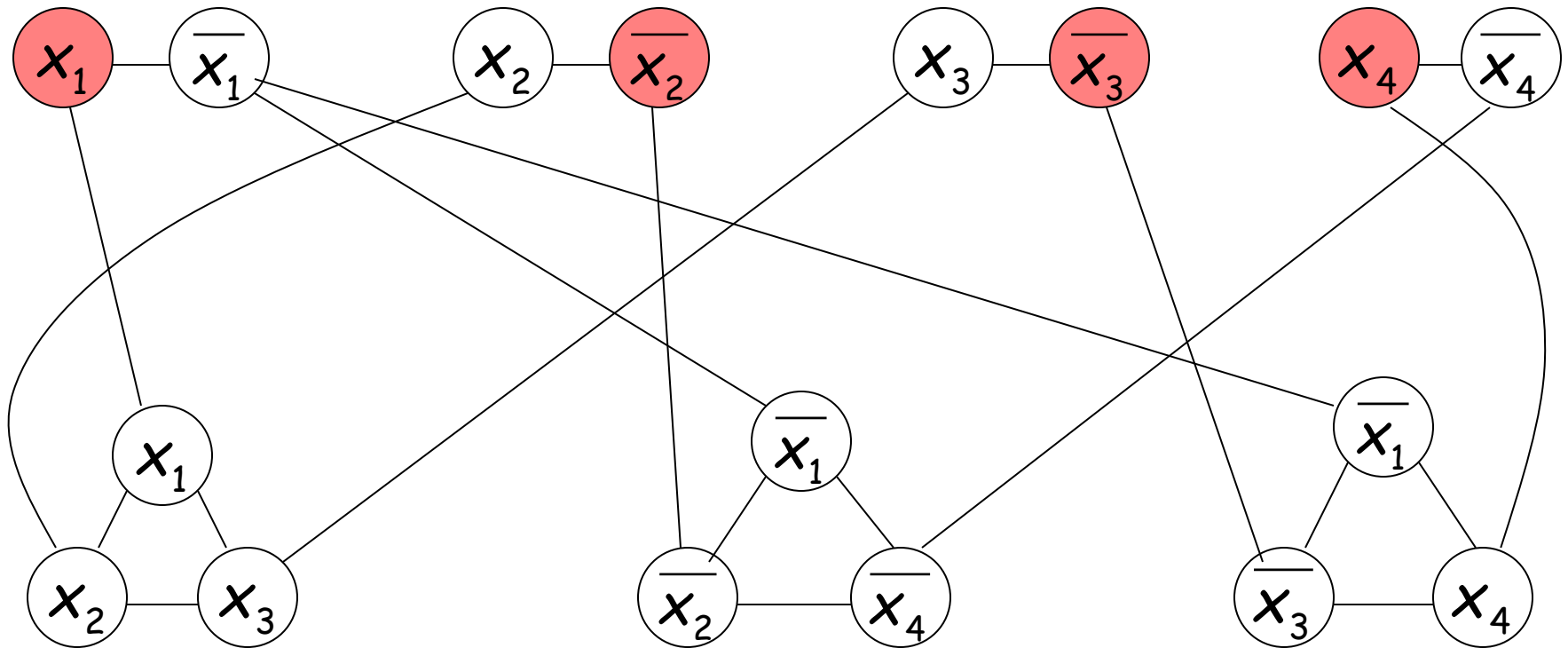
$$\varphi = (x_1 \vee x_2 \vee x_3) \wedge (\overline{x_1} \vee \overline{x_2} \vee \overline{x_4}) \wedge (\overline{x_1} \vee \overline{x_3} \vee x_4)$$

$$x_1 = 1$$

$$x_2 = 0$$

$$x_3 = 0$$

$$x_4 = 1$$



Put every satisfying literal in the cover

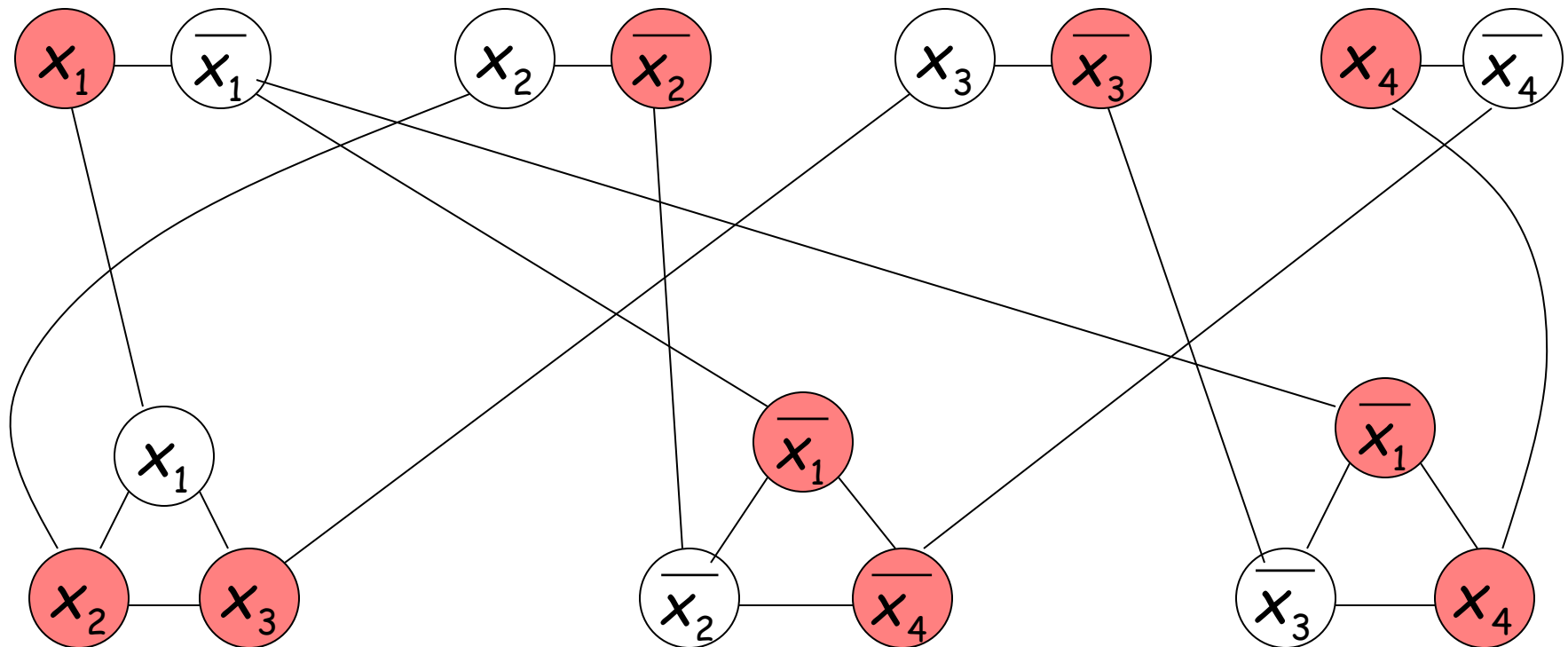
$$\varphi = (x_1 \vee x_2 \vee x_3) \wedge (\overline{x_1} \vee \overline{x_2} \vee \overline{x_4}) \wedge (\overline{x_1} \vee \overline{x_3} \vee x_4)$$

$$x_1 = 1$$

$$x_2 = 0$$

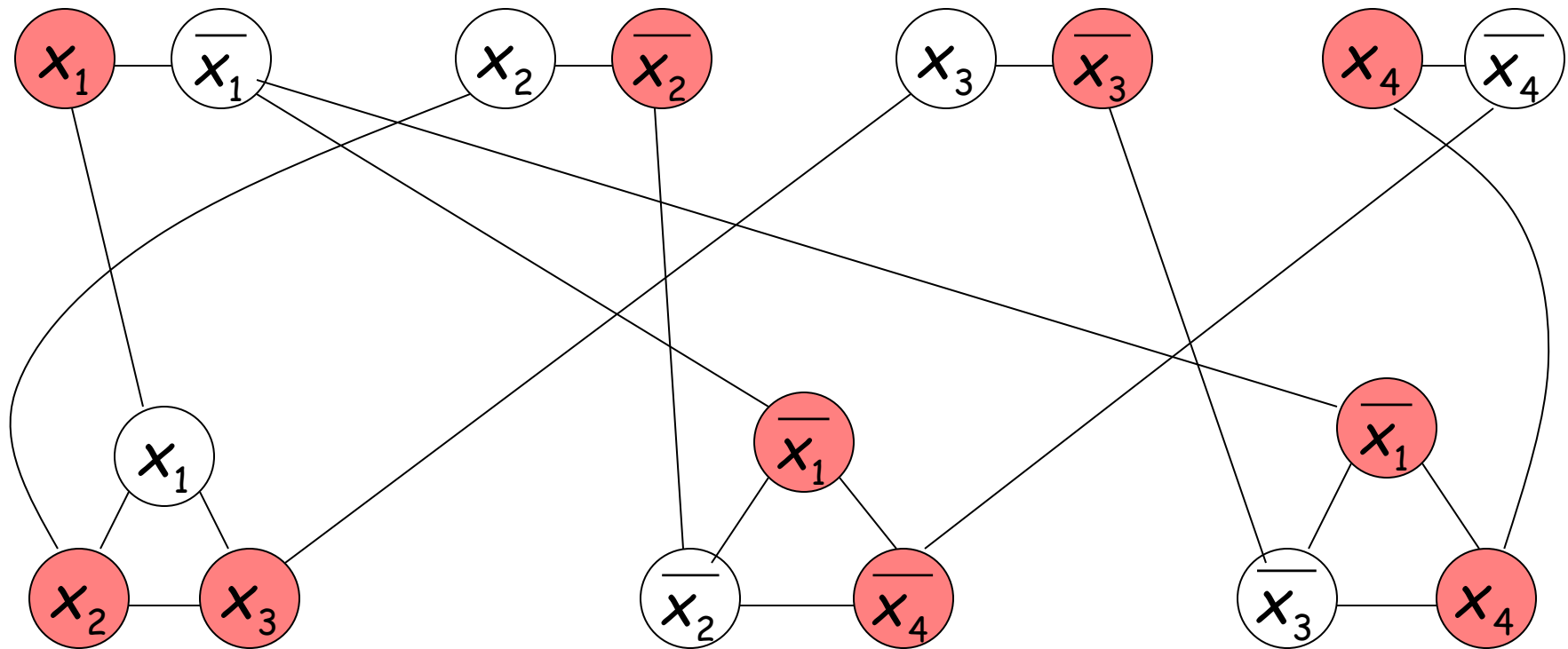
$$x_3 = 0$$

$$x_4 = 1$$



Select one satisfying literal in each clause gadget and include the remaining literals in the cover

This is a vertex cover since every edge is adjacent to a chosen node



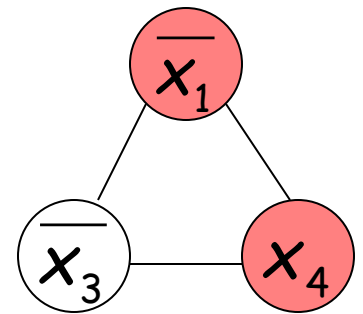
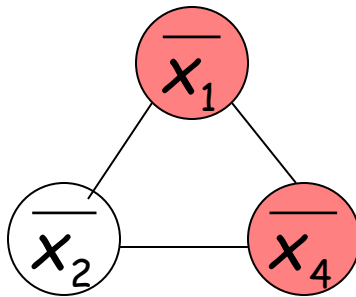
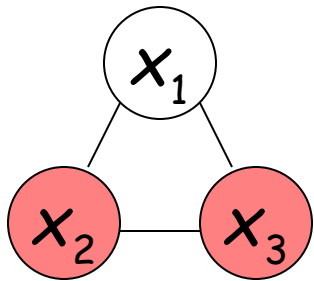


# Explanation for general case:

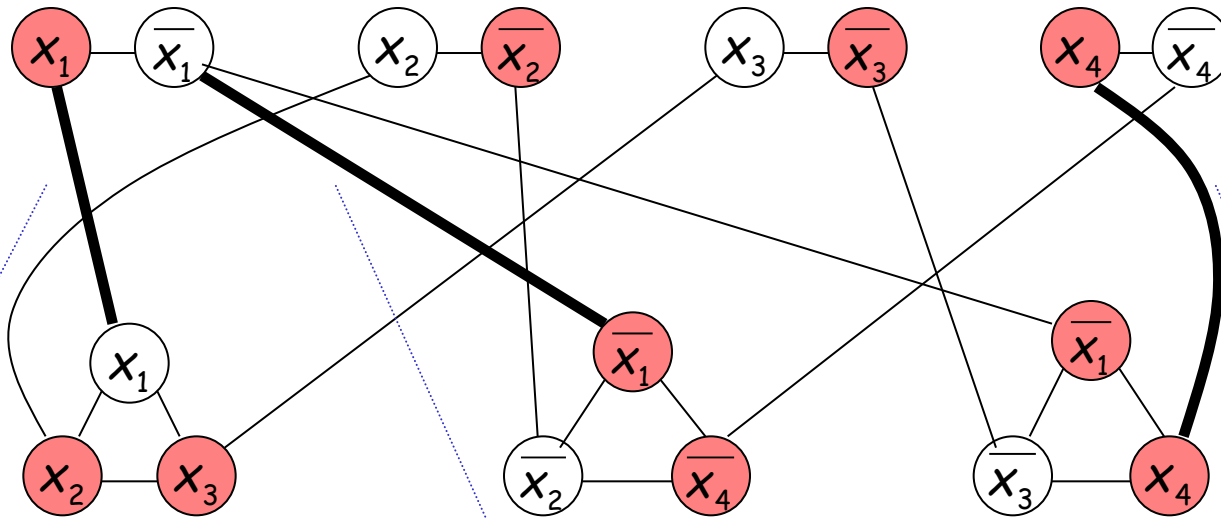


Edges in variable gadgets  
are incident to at least one node in cover

Edges in clause gadgets  
are incident to at least one node in cover,  
since two nodes are chosen in a clause gadget



Every edge connecting variable gadgets and clause gadgets is one of three types:



Type 1

Type 2

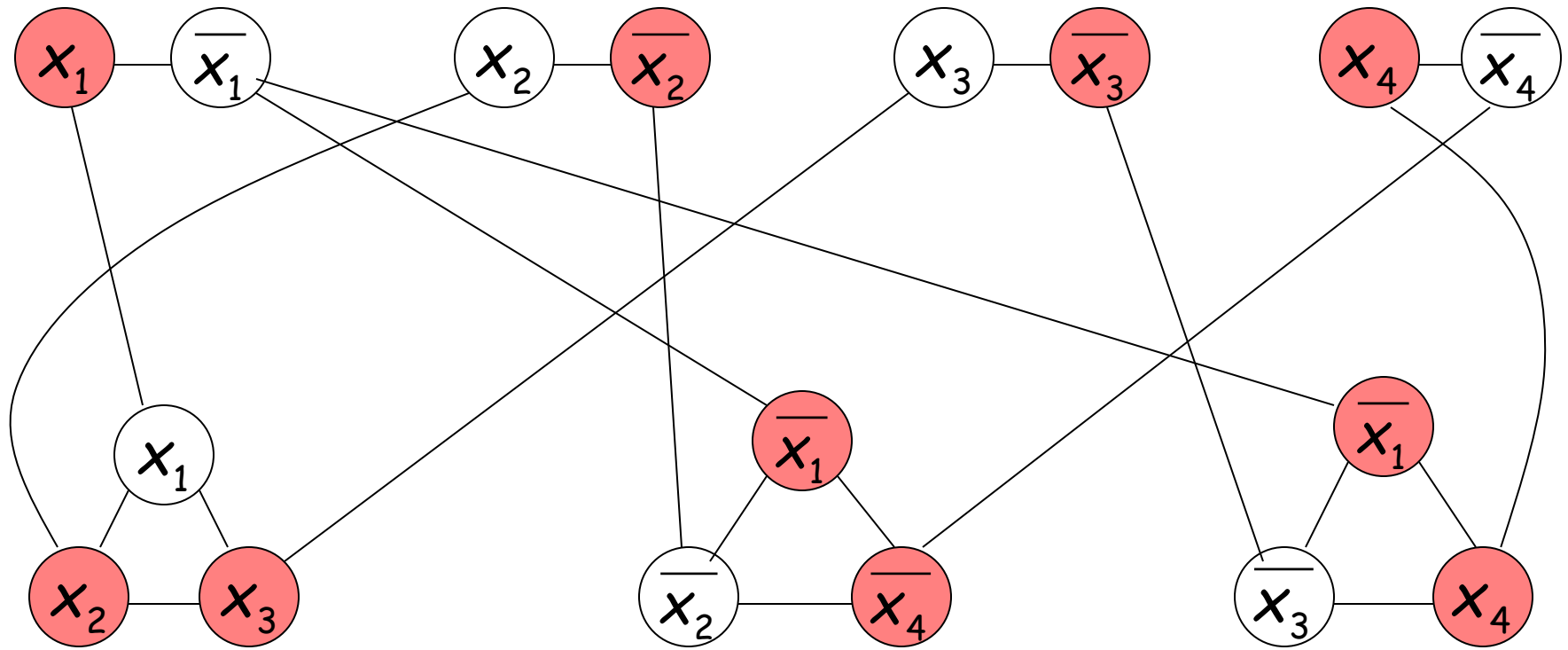
Type 3

All adjacent to nodes in cover

## Second direction of proof:

If graph  $G$  contains a vertex-cover  
of size  $k = m + 2$   
then formula  $\varphi$  is satisfiable

Example:



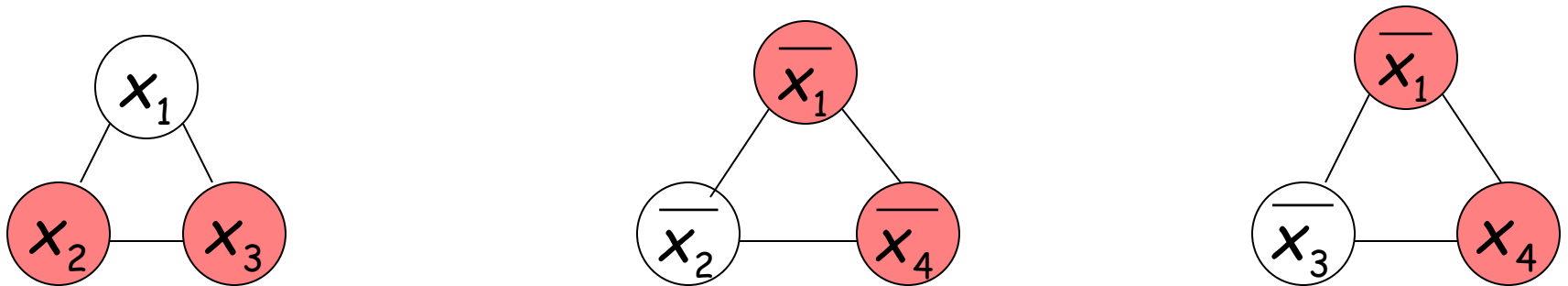
To include "internal" edges to gadgets,  
and satisfy  $k = m + 2l$

exactly one literal in each variable gadget is chosen



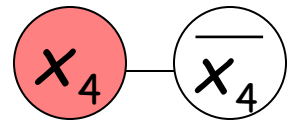
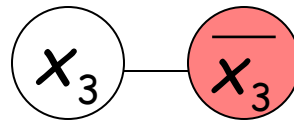
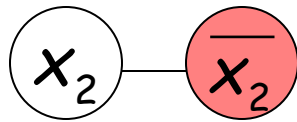
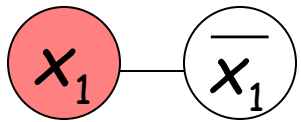
$m$  chosen out of  $2m$

exactly two nodes in each clause gadget is chosen



$2l$  chosen out of  $3l$

For the variable assignment choose the literals in the cover from variable gadgets



$$x_1 = 1$$

$$x_2 = 0$$

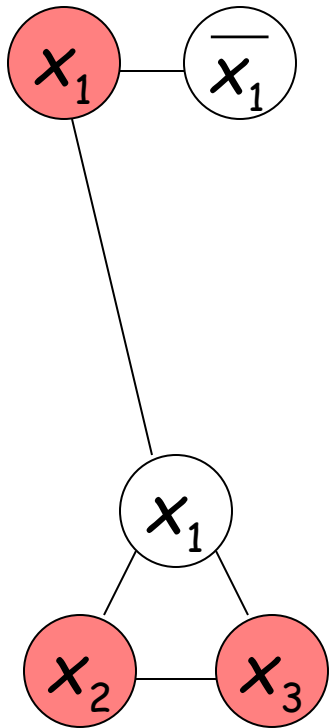
$$x_3 = 0$$

$$x_4 = 1$$

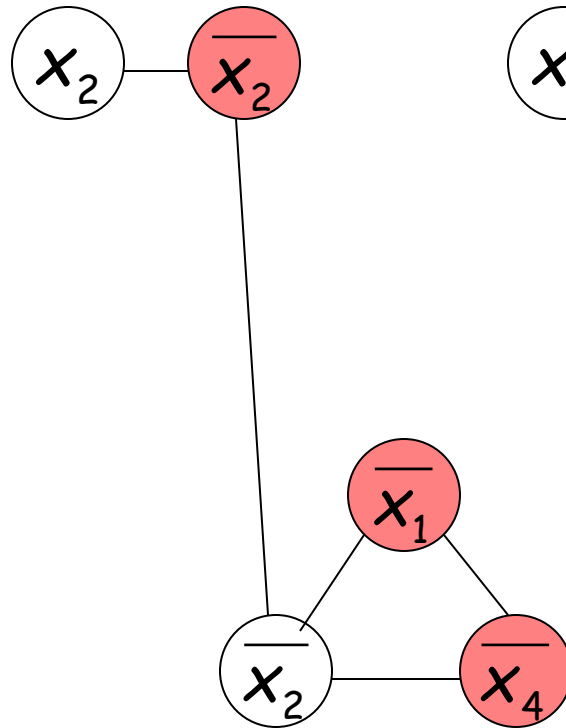
$$\varphi = (x_1 \vee x_2 \vee x_3) \wedge (\overline{x_1} \vee \overline{x_2} \vee \overline{x_4}) \wedge (\overline{x_1} \vee \overline{x_3} \vee x_4)$$

$\varphi = (x_1 \vee x_2 \vee x_3) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_4) \wedge (\bar{x}_1 \vee \bar{x}_3 \vee x_4)$   
is satisfied with

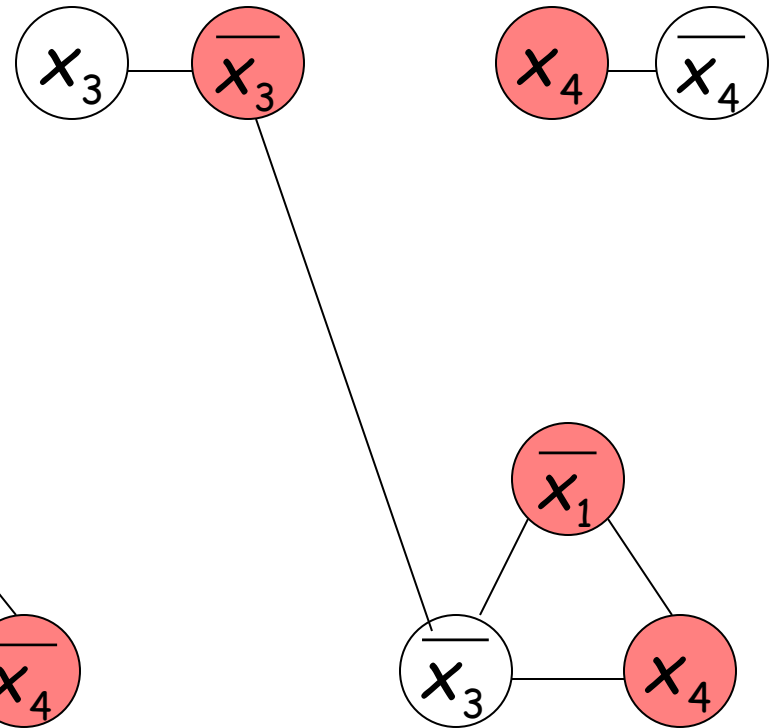
$$x_1 = 1$$



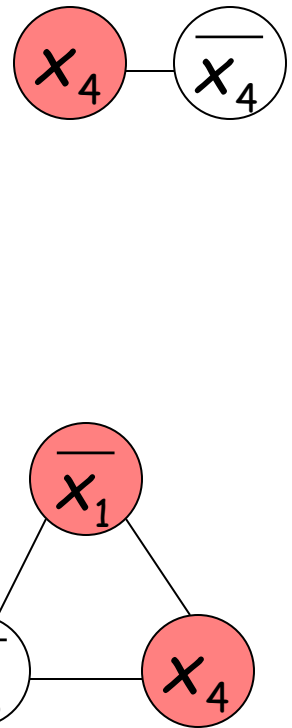
$$x_2 = 0$$



$$x_3 = 0$$



$$x_4 = 1$$



since the respective literals satisfy the clauses



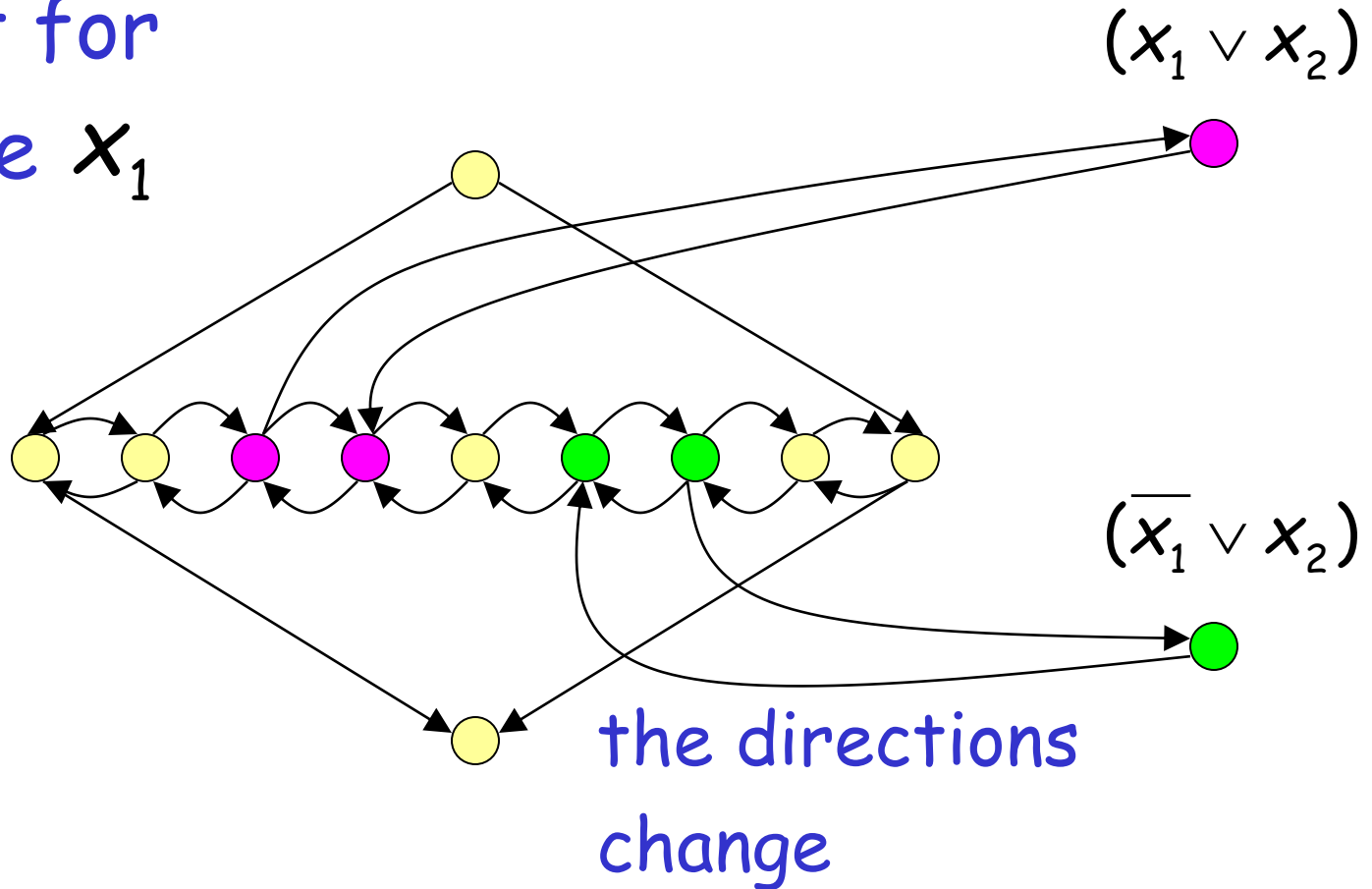
**Theorem:** HAMILTONIAN-PATH  
is NP-complete

**Proof:**

1. HAMILTONIAN-PATH is in NP  
Can be easily proven
2. We will reduce in polynomial time  
3CNF-SAT to HAMILTONIAN-PATH  
(NP-complete)

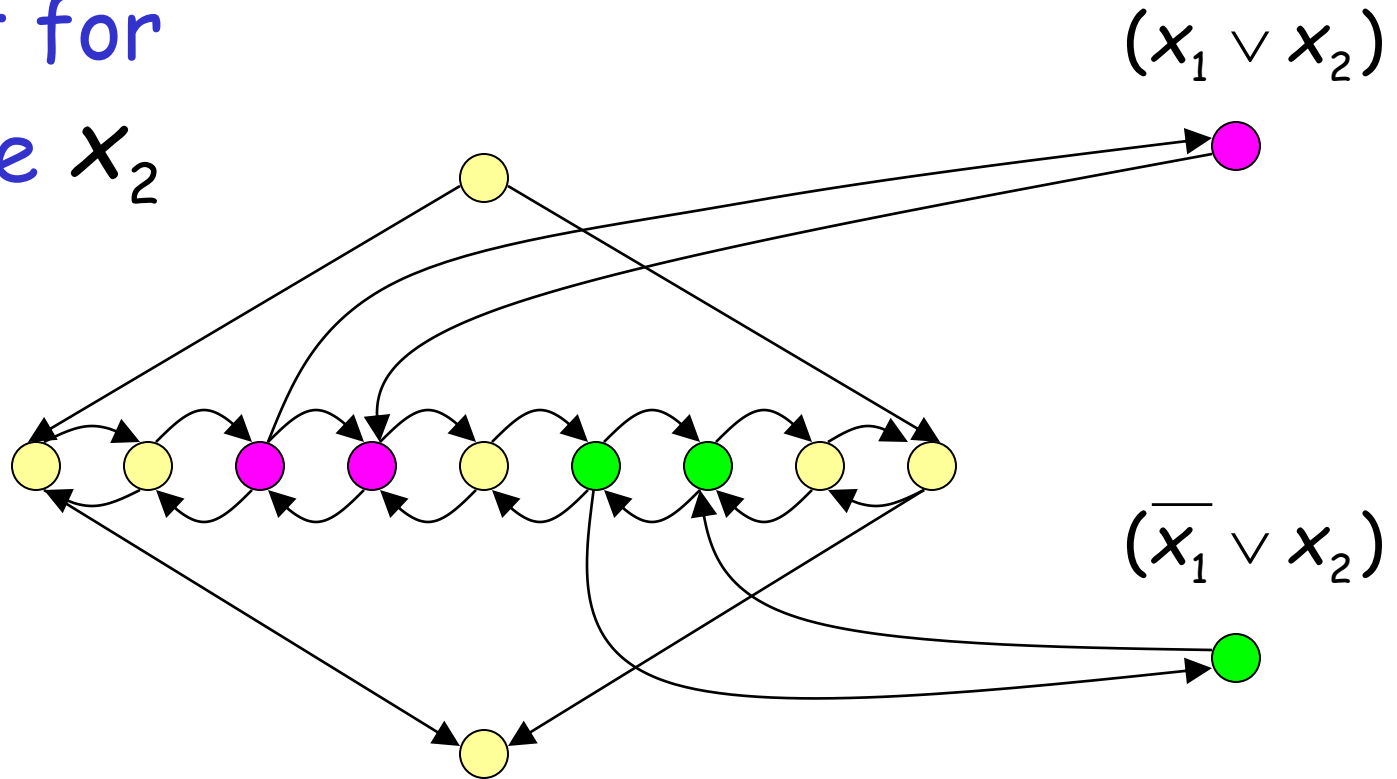
$$(x_1 \vee x_2) \wedge (\overline{x_1} \vee x_2)$$

Gadget for  
variable  $x_1$

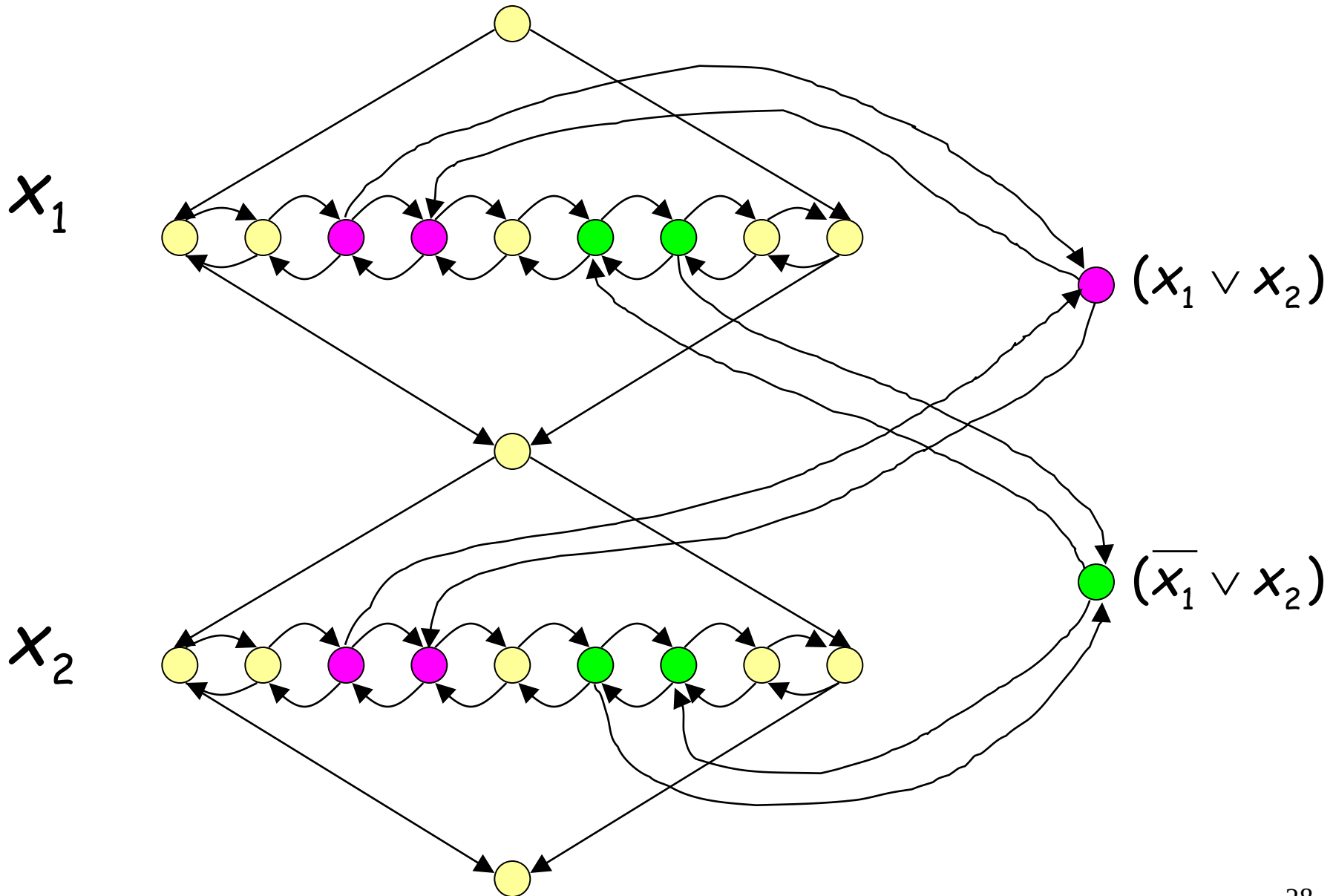


$$(x_1 \vee x_2) \wedge (\overline{x_1} \vee x_2)$$

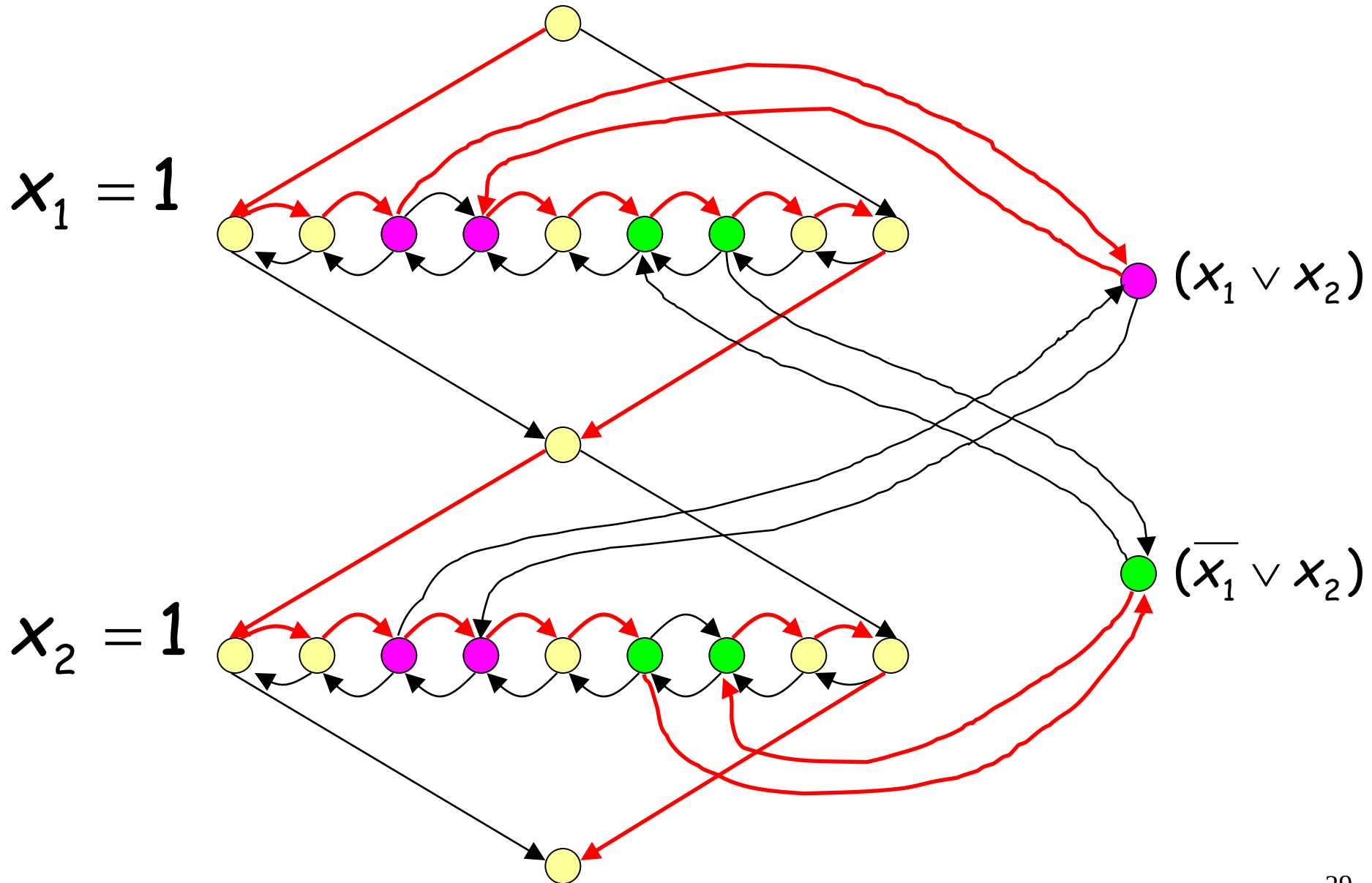
Gadget for  
variable  $x_2$



$$(x_1 \vee x_2) \wedge (\overline{x_1} \vee x_2)$$



$$(x_1 \vee x_2) \wedge (\overline{x_1} \vee x_2) = 1$$



$$(x_1 \vee x_2) \wedge (\overline{x_1} \vee x_2) = 1$$

